Bias–Variance General Concepts

1. Q: What is the **Bias–Variance Tradeoff** used to understand?
A: It's used to understand how model complexity affects prediction error — balancing underfitting and overfitting.

2. Q: What is the definition of **Bias** in the context of model prediction error?
A: Bias is the error introduced by approximating a real-world problem (which may be complex) by a simplified model.

3. Q: What is the definition of **Variance** in the context of prediction error?
A: Variance is the error caused by a model's sensitivity to fluctuations in the training data.

4. Q: Simpler models introduce **higher** what?
A: Bias.

5. Q: Why do simpler models typically introduce higher bias?
A: Because they cannot capture the true underlying patterns in the data, leading to systematic errors.

6. Q: The error in prediction due to a model's sensitivity to small fluctuations in the training dataset is called what?
A: Variance.

7. Q: The model capturing noise as a signal is a characteristic of what type of error?
A: High variance (overfitting).

8. Q: The tradeoff aims to find what kind of model?
A: A model with an optimal balance between bias and variance that minimizes total prediction error.

Bias, Variance, and Model Fitting

9. Q: High **Bias** leads to what model fitting problem?
A: Underfitting.

10. Q: High bias occurs because the model misses what?
A: The **underlying relationships** or patterns in the data.

11. Q: High **Variance** leads to what model fitting problem?
A: Overfitting.

12. Q: Why does high variance lead to poor performance on test data?
A: Because the model learns noise instead of the true signal and fails to generalize to unseen data.

Total Error Formulation (Conceptual)

13. Q: The **Total Error** for a model is composed of the squared value of what three main components?
A: Bias^2 + Variance + Irreducible Error (σ^2).

14. Q: What does the term y denote in the mathematical formulation?
A: The **true observed output** (dependent variable).

15. Q: What does the term f denote in the mathematical formulation?
A: The **true underlying function** that generates the data.

16. Q: What is the irreducible error, ε, assumed to be?
A: A random noise term with **mean 0** and **variance σ^2**.

17. Q: What is the estimate of the irreducible error that even a true model will incur?
A: Variance of ε (σ^2) — it cannot be reduced by any model.

Total Error Derivation

18. Q: What is the expression for the **Squared Error** being minimized?
A: $(y - \hat{f})^2$

19. Q: Recall that $y = f + \epsilon$; what is the final expression for $E[y^2]$?
A: $E[y^2] = E[f^2] + Var(\epsilon)$

20. Q: In the derivation, what is the expected value of the irreducible error, $E(\epsilon)$?
A: 0

21. Q: The variance of the irreducible error, σ^2, is equal to the expected value of what squared term?
A: $E[\epsilon^2]$

22. Q: Assuming ε and $\hat{f}$ are independent, what is the simplified expression for $E[2y\hat{f}]$?
A: $2E[f\hat{f}]$

23. Q: The final derived expression for the Squared Error is the sum of Bias², Variance, and what other term?
A: Irreducible Error (σ^2).

24. Q: Write the final mathematical identity for total squared error (Error).
A:
$$
E[(y - \hat{f})^2] = (\text{Bias}[\hat{f}])^2 + Var[\hat{f}] + \sigma^2
$$

Tradeoff Management

25. Q: To **decrease bias**, what should be done to model complexity?
A: Increase model complexity (make the model more flexible).

26. Q: What are two methods for decreasing variance mentioned in the document?
A: Regularization and **feature selection** (or data augmentation/ensemble averaging).

27. Q: Employing **regularization** is a method for decreasing which component of total error?
A: Variance.

28. Q: What technique related to features can be employed to decrease variance?
A: Reducing the number of features or using **feature selection/PCA.**

29. Q: On the Bias–Variance graph, which curve represents the **Total Error**?
A: The **U-shaped curve** formed by adding $Bias^2$ and Variance.

30. Q: On the Bias–Variance graph, what is the point where the Total Error curve is minimized?
A: The **optimal model complexity** or **sweet spot** between underfitting and overfitting.